

Mark Amgad

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SUMMARY

AI Engineering student with strong foundations in machine learning, algorithms, symbolic AI, and embedded systems. Experienced in building practical software and engineering-focused technical projects.

EDUCATION

B.Sc. Artificial Intelligence Engineering

2024 – Expected 2029

New Mansoura University

Egypt

- **Core Coursework:** OOP, Data Structures, Design & Analysis of Algorithms, Machine Learning, Knowledge-Based Systems, Discrete Mathematics

PROJECTS

Tech Diagnosis AI | Python, Symbolic AI, PySide6

- Built a knowledge-based expert system using symbolic AI with forward chaining inference engine for technical fault diagnosis
- Implemented certainty factor propagation to handle uncertain evidence and rank diagnoses by confidence level

Disease Prediction System | Python, Scikit-learn, Pandas

- Implemented a machine learning classification pipeline comparing Logistic Regression, Random Forest, SVM, and Neural Networks on clinical datasets
- Engineered preprocessing workflows including missing value handling, feature scaling, categorical encoding, and train-test split validation

Search Algorithm Visualizer | Python, Pygame, Algorithms

- Implemented and visualized pathfinding algorithms (A*, BFS, DFS, Dijkstra, Greedy Best-First) with step-by-step execution on grid-based mazes
- Added real-time metrics tracking showing nodes explored, path length, and execution time for algorithm comparison

Automated Smart Parking System (ASPS) | C++, Arduino, RFID, Sensors

- Built an embedded systems prototype using Arduino Uno with RFID card authentication and servo-controlled gate mechanism
- Integrated ultrasonic sensors for vehicle detection and real-time slot occupancy tracking with LED indicators

Bank Management System | C++, DSA, OOP

- Developed a console-based banking application implementing linked lists for account management and file I/O for persistent storage
- Designed object-oriented architecture with classes for accounts, transactions, and authentication using encapsulation and inheritance

EXPERIENCE

DecodeLabs

May 2025 – Jun. 2025

Data Analytics Intern

Remote / Virtual

- Performed data cleaning, transformation, and exploratory analysis on business datasets using Python and Pandas
- Automated repetitive preprocessing tasks with Python scripts to streamline workflow steps

TECHNICAL SKILLS

Languages : Python, C#, C++, SQL, Assembly 8086

AI & Data : Scikit-learn, Pandas, NumPy

Software Engineering : Data Structures & Algorithms, OOP, Problem Solving, Git, GitHub

Tools : Streamlit, Jupyter Notebook, Arduino IDE, AutoCAD, Excel

Embedded Systems : Arduino Uno, RFID Integration, Sensor Automation, Servo Control

CERTIFICATIONS

- Artificial Intelligence Ambassadors Program NT
- Introduction to Modern AI Cisco
- Data Analysis Training Program Creativa Innovation Hub
- AI Fundamentals IBM SkillsBuild
- AI for Work and Life University of North Florida
- ITIDA Gigs Freelancing Program

